

Syntax

- The **signature** $\mathcal{L}$ now consists of:
 - set of sorts, denoted by α, β
 - constants, denoted by a, b, c , each having a sort $c : \alpha$
 - function symbols, denoted by f, g , each having a type $\alpha_1 \times \dots \times \alpha_n \mapsto \alpha$
 - predicate symbols, denoted by p, q , each having a type $\alpha_1 \times \dots \times \alpha_n$

Quantifiers:

- Given a formula G , a variable x , then $\mathcal{Q}x : \alpha.A$ is a quantified formula with $\mathcal{Q} \in \{\forall, \exists\}$
- A formula is called quantifier-free (or ground) if it contains no occurrences of quantifiers

Semantics

- The **interpretation** I maps:
 - Each sort α to a non-empty set $I(\alpha)$, called domain of the sort
 - Each constant $c : \alpha$ to $c' \in I(\alpha)$
 - Each variable $x : \alpha$ to $x' \in I(\alpha)$
 - Each function symbol $f : \alpha_1 \times \dots \times \alpha_n \mapsto \alpha$ to a function $f' : I(\alpha_1) \times \dots \times I(\alpha_n) \mapsto I(\alpha)$
 - Each predicate symbol $p : \alpha_1 \times \dots \times \alpha_n$ to a relation $p' \subseteq I(\alpha_1) \times \dots \times I(\alpha_n)$

- Given I and a term t of a sort α , define the element $t' \in I(\alpha)$ as follows:

- For constants and variables $t' = I(t)$
- $f(t_1, \dots, t_n)' = f'(t_1', \dots, t_n')$

- Given I and a formula G , define the boolean value as follows:

- For an atomic formula $p(t_1, \dots, t_n)$, define $p(t_1, \dots, t_n)' = 1 = (t_1', \dots, t_n') \in p'$
- $G = (Q_1 \circ Q_2)' = 1$ if $Q_1' \circ Q_2'$ is satisfied

Quantifiers

- Given a set sequence of variables $\bar{x}$, two interpretations of the same signature are $\bar{x}$ -variant if they coincide on all symbols and variables not $\bar{x}$.
- $(\forall x.A)' = 1$ if for all $\bar{x}$ -variants I' of I we have $(A)' = 1$
- $(\exists x.A)' = 1$ if for some $\bar{x}$ -variant I' of I we have $(A)' = 1$

Axiomatic Interpretations

Restrict interpretations to those which satisfy the following set of formulas (so called axioms):

- reflexivity : $x = x$
- symmetry : $x = y \rightarrow y = x$
- transitivity : $x = y \wedge y = z \rightarrow x = z$
- congruence axioms for each function symbol f :

$$\left(\bigwedge_{i=1}^n x_i = y_i \right) \rightarrow f(x_1, \dots, x_n) = f(y_1, \dots, y_n)$$

- congruence axioms for each predicate symbol p :

$$\left(\bigwedge_{i=1}^n x_i = y_i \wedge p(x_1, \dots, x_n) \right) \rightarrow p(y_1, \dots, y_n)$$

Interpretation Classes

Restrict interpolations to those which satisfy certain properties

Example :

- Given the formula $f(x) = f(x+1) \wedge f(a) \neq f(b)$
- 1. We consider the class of interpretations I where I interprets $+$ and 1 in the standard way over the integers. The formula is satisfiable in:

$$I(a) = 1, I(b) = 2, I(x) = 3$$

$$f(x) = \begin{cases} 3 & \text{if } x = 4 \\ x & \text{else} \end{cases}$$

- 2. We consider the class of interpretations of example 1. but where $f(x) = f(x+1)$ is valid in I .

The formula is not satisfiable as $f(x) = f(x+1)$ would impose that f maps all variables to the same value, in which case $f(a) \neq f(b)$ can not hold.

Satisfiability and Interpretation - Examples

- Given the formula $x = y + 1$ over integers, where x, y are variables, $+$, 1 are interpreted in the standard way over integers and the interpretation $I(x) = I(y) = 0$. It then holds that:
 - $I \not\models x = y + 1$
 - $x = y + 1$ is satisfiable in I under the $\bar{x}$ -variant: $I'(x) = 1, I(y) = 0$
 - $x = y + 1$ is not valid in I , since $I \not\models x = y + 1$

- Given the formula $x \neq a \rightarrow z = b$, where $a : \alpha, b : \alpha$ are constants and $z : \alpha$ is a variable. This formula is valid under the class of interpretations I for which it holds that:

$$-D_a^I = \{0, 1\}, I(a) = 0, I(b) = 1$$

- Given the formula $y = a \rightarrow y = b$, where a, b are constants and y is a variable. This formula is valid under the class of interpretations I for which it holds that :
 - $I(a) = I(b)$

Example :

Given based on table to the right:

- $I \models F_1 \wedge F_2 \wedge F_3 \wedge F_4$
- $x, y, c_1, c_2, c_3, c_4 : \alpha ; f : \alpha \mapsto \alpha$

Let:

- $I(\alpha) := \mathbb{Z}, I(x), I(y), I(c_1), I(c_2), I(c_3), I(c_4) \in \alpha = \mathbb{Z}$
- $I(f) : \mathbb{Z} \mapsto \mathbb{Z}, I(a) : \mathbb{Z} \mapsto \mathbb{Z}$ (which means, that array a is interpreted as a function)

Set:

- $I(c_1) = 4, I(c_4) = 3, I(x) = I(c_2) = 0, I(y) = I(x) + 2 = 2$
- $I(a)(i) := 0$ forall $i \in \mathbb{Z}$

$$-I(f)(v) = \begin{cases} 1 & v = 4 \\ 0 & \text{else} \end{cases}$$

Theory

A theory is defined as

- a signature Σ_T
- a set of formulas $\mathcal{A}_T$ over Σ_T , called axioms of $\mathcal{T}$

Unless specified otherwise, theories are quantifier-free

- An interpretation in which $\mathcal{A}_T$ are valid is called a $\mathcal{T}$ -interpretation
- If a formula G is valid in every $\mathcal{T}$ -interpretation, then G is $\mathcal{T}$ -valid (denoted by $\mathcal{T} \models G$)

Theory of equality $\mathcal{T}_E$:

- $\Sigma_E = \{a, b, \dots, f, g, \dots, p, \dots\}$
- $A_E = \left\{ \begin{array}{l} x = x, x = y \rightarrow y = x, x = y \wedge y = z \rightarrow x = z, \\ \bigwedge_{i=1}^n y_i = x_i \rightarrow f(x_1, \dots, x_n) = f(y_1, \dots, y_n), \\ \bigwedge_{i=1}^n y_i = x_i \wedge p(x_1, \dots, x_n) \rightarrow p(y_1, \dots, y_n) \end{array} \right\}$

A binary relation R over a set S is called a congruence relation, if:

- R is reflexive, transitive, symmetric and
- $\bigwedge_{i=1}^n y_i R x_i \rightarrow f(x_1, \dots, x_n) R f(y_1, \dots, y_n)$

The congruence class of $t \in S$ under R is $[t]_R = \{t' \in S \mid t R t'\}$,

for which holds:

- $\bigcup_{[t]_R} [t]_R = S$
- $[t_1]_R \neq [t_2]_R \rightarrow [t_1]_R \cap [t_2]_R = \emptyset$

The congruence closure R^c of R is the relation for which holds:

- $R \subseteq R^c$
- $\forall R' : R \subseteq R' \rightarrow (R^c = R' \vee R^c \subseteq R')$

Congruence Closure Algorithm for $\mathcal{T}_E$

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procedure CongruenceClosure( $F$ )
input:  $F$  is  $s_1 = t_1 \wedge \dots \wedge s_n = t_n \wedge s_{n+1} \neq t_{n+1} \wedge \dots \wedge s_m \neq t_m$ 
output: satisfiable or unsatisfiable
parameters: function subterm.set
begin
   $S := \text{subterm.set}(F)$ 
   $R := \{sRs \mid s \in S\}$  and  $[s]_R := \{s\}$ , defining the partition  $\{[s]_R \mid s \in S\}$ 
  for every  $s_i = t_i$  in  $F$ , merge  $[s_i]_R$  and  $[t_i]_R$  by
  forming the union  $[s_i]_R \cup [t_i]_R$ 
  propagate the new congruences that arise in this union (function congruence)
  if  $s_i R t_i$  for any  $i \in \{n+1, \dots, m\}$  then return unsatisfiable
else return satisfiable end

```

Example

Given the formula $F : f(a, b) = a \wedge f(f(a, b), b) \neq a$

We proceed as follows:

- 1. Subterm Set $S_F = \{a, b, f(a, b), f(f(a, b), b)\}$
- 2. Initial partition $P = \{\{a\}, \{b\}, \{f(a, b)\}, \{f(f(a, b), b)\}\}$

- 3. Given $f(a, b)Ra$, we merge the partition :

$$P = \{\{a, f(a, b)\}, \{b\}, \{f(f(a, b), b)\}\}$$

- 4. Given $f(a, b)Ra \wedge bRb \rightarrow f(f(a, b), b)Rf(a, b)$:

$$P = \{\{a, f(a, b), f(f(a, b), b)\}, \{b\}\}$$

- 5. Due to the clause $f(f(a, b), b) \neq a$, formula F is $\mathcal{T}_E$ -unsatisfiable, since $f(f(a, b), b) \in [a]_R \iff f(f(a, b), b) R a \in R$, but the formula states, that $f(f(a, b), b) \neq a$

DPLL(T) - For Non-Unit Clauses in $\mathcal{T}_E$

Given a formula $G := (a = b \vee a = c) \wedge \dots$ then G contains a non-unit clause and can thus not be proven with congruence closure.

Instead, use DPLL $_T$:

```

procedure DPLL $_T$ ( $F$ )
input: a CNF formula  $F$  in the decidable theory  $\mathcal{T}$ 
output: satisfiable or unsatisfiable
parameters: functions propositional.abstraction, decision.procedure $_T$ 
begin
   $S := \text{propositional.abstraction}(F)$ 
  while true do
    if DPLL( $S$ ) is unsatisfiable then return unsatisfiable
    if DPLL( $S$ ) is satisfiable then
       $L := \{L_1, \dots, L_n\}$ , s.t. for the model  $I$  of  $I$  we have  $I \models S$ 
      if decision.procedure $_T$ ( $L$ ) is satisfiable then return satisfiable
      if decision.procedure $_T$ ( $L$ ) is unsatisfiable then
         $S := S \cup \{\neg L_1 \vee \dots \vee \neg L_n\}$ 
      end do
  end

```

Example

Given the formula F :

$$(a = b \vee a = c) \wedge f(a) = a \wedge f(f(b)) \neq a \wedge f(c) \neq a$$

- 1. Add propositional symbol for every theory atom:

$$p_1 : a = b, p_2 : a = c, p_3 : f(a) = a,$$

$$p_4 : f(f(b)) \neq a, p_5 : f(c) = a$$

$$F_P = (p_1 \vee p_2) \wedge p_3 \wedge \neg p_4 \wedge \neg p_5$$

- 2. If F_P is unsat, then so is F

- 3. If F_P is sat, then we have literals $L_1, \dots, L_n$

of I for which holds: $I \models F_P$

$$\text{e.g. } p_1, p_2, p_3, \bar{p}_4, \bar{p}_5$$

- 4. Use clauses corresponding to literals and check with congruence closure

$$F' = a = b \wedge a \neq c \wedge f(a)$$

$$\stackrel{\text{simp}}{\implies} \bar{p}_5$$

- 5. If F' is $\mathcal{T}$ -sat, terminate, else define $F_P = F_P \wedge (\neg L_1 \vee \dots \vee \neg L_2)$ and go to 2.

Theory of arrays $\mathcal{T}_A$:

- $\Sigma_A = \{\text{read}, \text{write}\}$
- where read is a binary function and write is ternary function
- $A_A = A_E \cup \left\{ \begin{array}{l} x = y \rightarrow \text{read}(\text{write}(A, x, v), y) = v, \quad \text{r-o-w}_1 \\ x \neq y \rightarrow \text{read}(\text{write}(A, x, v), y) = \text{read}(A, y) \quad \text{r-o-w}_2 \end{array} \right\}$
- $\text{read}(A, x)$ denotes the value of array A at pos. x
- $\text{write}(A, x, v)$ denotes the modified array A with value v at pos. x

Given a function F with write -terms $\text{read}(\text{write}(A, x, v), y)$, $F[F']$ denotes the formula where the write -term has been replaced with F'

Congruence Closure Algorithm for $\mathcal{T}_A$

$\mathcal{T}_A$ -satisfiability of F is reduced to $\mathcal{T}_E$ -satisfiability:

- If F contains write -terms $\text{read}(\text{write}(A, x, v), y)$, use r-o-w axioms:

- 1. $F_1 : x = y \wedge F[v]$
- 2. $F_2 : x \neq y \wedge F[\text{read}(A, y)]$
- If both are unsat, then F is unsat

- If F doesn't contain write -terms:

- Use fresh function symbol f_A for array A
- Replace all $\text{read}(A, x)$ with $f_A(x)$

Return $\mathcal{T}_E$ satisfiability of modified formula

Combination of Theories

Given:

- Given a set of theories $\mathcal{T}_1, \dots, \mathcal{T}_n$ in disjoint $\Sigma_1, \dots, \Sigma_n$ and its axioms $\bigcup_{i=1}^n \mathcal{A}_{\mathcal{T}_i}$

- Quantifier-free conjunction F of literals in $\bigcup_{i=1}^n \Sigma_i$

Checking if F is satisfiable in a $\mathcal{T}$:

- Transform $F : \bigwedge_{i=1}^n F_i$ with F_i being a $\mathcal{T}_i$ formula, where $F_1, \dots, F_n$ are linked by variables

Example:

Given $\mathcal{T}_A, \mathcal{T}_E$ and $F : (y = x + 2) \wedge (f(\text{read}(\text{write}(a, x, 3), y - 2)) \neq f(y - x + 1))$, we split:

$F : (y = x + 2) \wedge f(c_1) \neq f(c_1)$

Shared equalities are derived e.g

$$-y = x + 2 \wedge c_2 = y - 2$$

$$\implies c_2 = x$$

$$-c_4 = \text{read}(\text{write}(a, x, c_3), c_2) \wedge c_2 = c_x$$

$$\stackrel{\text{r-o-w}_1 \text{ axiom}}{\implies} c_4 = c_3$$

$$\implies c_4 = c_3$$

$$-c_4 = \text{read}(\text{write}(a, x, c_3), c_2) \wedge c_2 = c_x$$

$$\implies c_4 = c_3$$

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